

Literature-Already-Answered Packet

Weak compact sets being super weakly compact forces weak sequential completeness

Status: literature already answered, not a new automatic-discovery result

Source and later answer

The source paper is:

arXiv:2408.01135, Zdenek Silber, *Weak compactness in Lipschitz-free spaces over super-reflexive spaces.*

The local copy is included as `source_paper.pdf`.

The later supporting paper is:

arXiv:2501.16855, Zdenek Silber, *On weak sequential completeness of spaces where weakly compact sets are super weakly compact.*

The local copy is included as `supporting_paper_2501.16855.pdf`.

Open question location

The following full-width crop is taken from page 11 of the source PDF. It shows Question 1 in the open-problems section and the author's James-space candidate remark.

4.2. **Open problems.** Note that all the examples of Banach spaces in which every relatively weakly compact set is relatively super weakly compact mentioned above are weakly sequentially complete. This inspires the following question:

Question 1. *Is there a Banach space in which every relatively weakly compact set is relatively super weakly compact but which is not weakly sequentially complete?*

A natural candidate to test for Question 1 is the James's space or some of its variants.

The next problem is concerned about super-variants of some properties studied in this paper.

Question 2. *Is the space $\mathcal{T}(X)$ super weakly sequentially complete? Does it have super weak*

Original question

Silber asks whether there exists a Banach space in which every relatively weakly compact set is relatively super weakly compact, but which is not weakly sequentially complete [1].

Later literature answer

The later paper explicitly cites the original question and states that it gives a negative answer. The next crop is from page 1 of the supporting PDF.

arXiv:2501.16855v

$C(K, \mathbb{R}^n)$ [11, Remark 3.4., Theorem 3.6.], or Lipschitz-free spaces over some special kinds of Carnot-Carathéodory metric spaces [2, Theorem 3.7.].

In [11], the so-called property (R) was investigated – a Banach space X has property (R) if and only if every weakly precompact subset of X is relatively super weakly compact. It can be easily seen that property (R) of X is equivalent to the fact that X is weakly sequentially complete and every relatively weakly compact subset of X is relatively super weakly compact. As all the known Banach spaces where every weakly compact set is super weakly compact (see the list in the previous paragraph) are also weakly sequentially complete, the following question was asked [11, Question 1]:

Question 1. *Is there a Banach space in which every weakly compact set is super weakly compact but which is not weakly sequentially complete?*

In this paper we give a negative answer to this question. Our method is based on the notion of a summing sequence – a seminormalised basic sequence on which we can define the summing functional. This notion was used by Rosenthal under the name wide-(s) sequence [10] (the same notion was originally used under the name P^* sequence by Singer [12]). By a result from [10],

The decisive statement is Theorem 2.8 of the later paper.

$(s_n)_{n \in \mathbb{N}}$ clearly satisfies the conclusion of the proposition. □

Theorem 2.8. *Let X be a Banach space where relatively weakly compact sets are relatively super weakly compact. Then X is weakly sequentially complete.*

Proof. Suppose that X is not weakly sequentially complete. Then there is a nontrivial weakly Cauchy sequence $(x_n)_{n \in \mathbb{N}}$ in X . Using Proposition 2.5 we can without loss of generality assume

Idea of the proof

This packet does not claim an original proof. The identification is direct: the source asks whether the “weakly compact sets are super weakly compact” property can occur without weak sequential completeness, while the supporting paper proves that the property forces weak sequential completeness.

The supporting proof starts from a nontrivial weakly Cauchy sequence. A Rosenthal summing subsequence produces uniformly bounded functionals with a triangular pattern. Differences of the weakly Cauchy sequence form a relatively weakly compact convex set, and the triangular functionals witness the finite tree obstruction from the Lancien–Raja characterization of super weak compactness. Thus any failure of weak sequential completeness manufactures a relatively weakly compact set that is not relatively super weakly compact [2].

Theorem

Theorem. *There is no Banach space in which every relatively weakly compact set is relatively super weakly compact but which is not weakly sequentially complete.*

Proof. This is exactly the contrapositive of Theorem 2.8 in arXiv:2501.16855: if relatively weakly compact sets in a Banach space X are relatively super weakly compact, then X is weakly sequentially complete [2]. $\square$

Classification

This packet belongs in `solutions/literature_already_answered/`. The later paper explicitly cites the original problem source and says it gives a negative answer to the question. Therefore this is a successful status/retrieval hit, not an original automatic-discovery proof.

Bounded literature and duplicate check

Before packaging, the run indexes were searched:

- index files: `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`;
- proof-gap index: `proof_gaps/index.tsv`;
- identifiers: 2408.01135 and 2501.16855;
- phrases: the title phrase, “weakly compact sets are super weakly compact”, “weakly sequentially complete”, and “Question R-noWSC”.

No existing packet or attempt covered this exact target. External search found arXiv:2501.16855, whose abstract and introduction explicitly identify the answer to Silber’s 2024 question.

Reproducibility

The screenshot crops were generated from the included PDFs by

`code/crop_evidence.py`.

All LaTeX build products are placed in `tmp/`. The final rendered packet is `solution_packet.pdf`.

References

- [1] Z. Silber, *Weak compactness in Lipschitz-free spaces over superreflexive spaces*, arXiv:2408.01135, 2024.

- [2] Z. Silber, *On weak sequential completeness of spaces where weakly compact sets are super weakly compact*, arXiv:2501.16855, 2025.